

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279335

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

MASUDA; Shinichi et al.

SEMICONDUCTOR MODULE AND POWER ELECTRONICS DEVICE

Abstract

A semiconductor module includes insulated circuit boards and a heat dissipation base having a rectangular top surface on which the plurality of insulated circuit boards is disposed, and having four fastener holes that penetrate through the heat dissipation base in a thickness direction at respective ones of corners of the top surface. The heat dissipation base further includes a pair of fixing portions each having a fixing hole penetrating therethrough in the thickness direction. The pair of fixing portions protruding, in a plan view of the semiconductor module, outwardly respectively from central parts of a pair of first edges of the top surface. The pair of first edges oppose each other and extend in a longer-side direction of the top surface.

Inventors: MASUDA; Shinichi (Matsumoto-city, JP), KAWABATA; Junya (Matsumoto-city, JP)

Applicant: FUJI ELECTRIC CO., LTD. (Kawasaki-shi, JP)

Family ID: 96831843

Assignee: FUJI ELECTRIC CO., LTD. (Kawasaki-shi, JP)

Appl. No.: 19/041659

Filed: January 30, 2025

Foreign Application Priority Data

JP	2024-029795	Feb. 29, 2024
----	-------------	---------------

Publication Classification

Int. Cl.: H01L23/40 (20060101); H01L23/055 (20060101); H01L23/538 (20060101);
H01L25/07 (20060101)

U.S. Cl.:

CPC **H01L23/4006** (20130101); **H01L23/5385** (20130101); **H01L25/072** (20130101);
H01L23/055 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based upon and claims the benefit of priority of the prior Japanese Patent Application No. 2024-029795, filed on Feb. 29, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The embodiments discussed herein relate to a semiconductor module and a power electronics device.

2. Background of the Related Art

[0003] A semiconductor module includes semiconductor chips (power devices), insulated circuit boards on which the semiconductor chips are disposed, and a heat dissipation plate on which the multiple insulated circuit boards are disposed (see, for example, the following patent references (1) to (8)). A plurality of such semiconductor modules may be attached in parallel to a heat dissipation base (see, for example, the following patent references (1), (2), (6), and (8)). In this case, the semiconductor modules are attached to the heat dissipation base via joining members (for example, thermal grease, solder, or adhesive materials) (see, for example, the following patent references (1) and (6)). [0004] (1) Japanese Laid-open Patent Publication No. 2013-004880 [0005] (2) Japanese Laid-open Patent Publication No. 2004-319992 [0006] (3) Japanese Laid-open Patent Publication No. 2007-059902 [0007] (4) Japanese Laid-open Patent Publication No. H04-229502 [0008] (5) Japanese Laid-open Patent Publication No. 2012-043915 [0009] (6) Japanese Patent No. 7258269 [0010] (7) Japanese Laid-open Patent Publication No. 2023-085765 [0011] (8) International Publication Pamphlet No. WO 2013/145619

SUMMARY OF THE INVENTION

[0012] According to an aspect, there is provided a semiconductor module including a plurality of insulated circuit boards; and a heat dissipation base having a rectangular top surface on which the plurality of insulated circuit boards is disposed, and having four fastener holes that penetrate through the heat dissipation base in a thickness direction at respective ones of corners of the top surface, wherein: the heat dissipation base further includes a pair of fixing portions each having a fixing hole penetrating therethrough in the thickness direction, the pair of fixing portions protruding, in a plan view of the semiconductor module, outwardly respectively from central parts of a pair of first edges of the top surface, which opposes each other and extends in a longer-side direction of the top surface.

[0013] The object and advantages of the invention will be realized and attained by means of the elements and combinations particularly pointed out in the claims.

[0014] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are not restrictive of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] FIG. 1 is a plan view of a semiconductor module according to a first embodiment;

[0016] FIG. **2** is a side view of the semiconductor module according to the first embodiment;
[0017] FIG. **3** is a cross-sectional side view of the semiconductor module according to the first embodiment;
[0018] FIG. **4** is a plan view of a heat dissipation base with insulated circuit boards disposed thereon, included in the semiconductor module of the first embodiment;
[0019] FIG. **5** is a side view of the semiconductor module attached to a cooling module according to the first embodiment;
[0020] FIG. **6** is a side view of a semiconductor module attached to a cooling module (at room temperature) according to a reference example;
[0021] FIG. **7** is a side view of the semiconductor module attached to the cooling module (at high temperature) according to the reference example;
[0022] FIG. **8** is a plan view of a power electronics device including the semiconductor modules of the first embodiment;
[0023] FIG. **9** is a plan view of a semiconductor module according to a second embodiment;
[0024] FIG. **10** is a plan view of a semiconductor module according to a third embodiment; and
[0025] FIG. **11** is a plan view of a power electronics device including the semiconductor modules of the third embodiment.

DETAILED DESCRIPTION OF THE INVENTION

[0026] Several embodiments will be described below with reference to the accompanying drawings. In the following, the terms “front surface” and “top surface” refer to the X-Y plane facing upward (in the +Z direction) in semiconductor modules and power electronics devices of the drawings. Similarly, the term “upper” refers to the upward direction (the +Z direction) of the illustrated semiconductor modules and power electronics devices. On the other hand, the terms “back surface” and “bottom surface” refer to the X-Y plane facing downward (in the -Z direction) in the illustrated semiconductor modules and power electronics devices. Similarly, the term “lower” refers to the downward direction (the -Z direction) of the illustrated semiconductor modules and power electronics devices. These terms have the same orientational relationships as described above in all the drawings if needed. The terms “front surface”, “top surface”, “upper”, “back surface”, “bottom surface”, “lower”, and “lateral surface” are simply expedient expressions used to specify relative positional relationships, and are not intended to limit the technical ideas of the embodiments described herein. For example, the terms “upper” and “lower” do not necessarily imply the vertical direction to the ground surface. That is, the “upper” and “lower” directions are not defined in relation to the direction of the gravitational force. In addition, the term “major component” in the following refers to a constituent having a concentration equal to 80 vol % or higher.

(a) First Embodiment

[0027] A semiconductor module **10** according to a first embodiment is described with reference to FIGS. **1** to **3**. FIG. **1** is a plan view of a semiconductor module according to the first embodiment. FIG. **2** is a side view of the semiconductor module according to the first embodiment. FIG. **3** is a cross-sectional side view of the semiconductor module according to the first embodiment. The cross-sectional side view of FIG. **3** is taken along dash-dotted line I-I in FIG. **1**. Note that FIG. **3** omits the illustration of wires.

[0028] The semiconductor module **10** includes a heat

[0029] dissipation base **30** disposed on the back surface and a case **40** disposed on the heat dissipation base **30** and covering the sides of the heat dissipation base **30**. The semiconductor module **10** includes components housed in a housing area **41g** surrounded by the heat dissipation base **30** and the case **40**, and the components in the housing area **41g** are sealed with a sealing member **47**. The components included in the housing area **41g** are, for example, insulated circuit boards, semiconductor chips disposed on the insulated circuit boards, and wires connecting these. FIG. **3** depicts insulated circuit boards **20a** and **20b**, as part of the insulated circuit boards, and

semiconductor chips **24**. The semiconductor module **10** includes external connection terminals **43** to **45**.

[0030] The heat dissipation base **30** is a plate-shaped member in plan view. The outer shape of the heat dissipation base **30** may correspond to the outer shape of the case **40**, except for fixing portions **32a** and **32b** to be described later. The heat dissipation base **30** may have R- or C-chamfered corners. The heat dissipation base **30** is made of a metal having excellent heat dissipation properties. Such a metal is, for example, copper, aluminum, or an alloy containing at least one of these. Plating may be applied to coat the surfaces of the heat dissipation base **30** in order to provide improved corrosion resistance. In this case, a material used for plating is, for example, nickel, a nickel-phosphorus alloy, or a nickel-boron alloy. Details of the heat dissipation base **30** will be described later.

[0031] To the top surface of the heat dissipation base **30**, insulated circuit boards **20a** to **20d** and insulated circuit boards **20e** and **20f** (see FIG. 4) are joined via joining members (not illustrated). The semiconductor chips **24** are joined to the insulated circuit boards **20a** to **20d** via joining members. Note that the insulated circuit boards **20a** to **20d** each include an insulating plate **21**, a metal plate **22** disposed on the back surface of the insulating plate **21**, and a conductive circuit pattern **23** disposed on the front surface of the insulating plate **21**. The insulated circuit boards **20e** and **20f** (secondary insulated circuit boards) are similarly configured.

[0032] The joining members are, for example, solder. The solder used is lead-free solder. The lead-free solder contains, as a major component, at least one alloy selected from a tin-silver-copper alloy, a tin-zinc-bismuth alloy, a tin-copper alloy, and a tin-silver-indium-bismuth alloy, for example. Further, the solder may include an additive, such as nickel, germanium, cobalt, antimony, or silicon. The inclusion of the additive increases wettability, brightness, and bond strength of the solder, which results in improved reliability. The joining members may be sintered compacts instead of solder. The material of the sintered compacts used in joining is, for example, a powder of silver, iron, copper, aluminum, titanium, nickel, tungsten, or molybdenum.

[0033] The insulating plates **21** are each rectangular in plan view. The insulating plates **21** may have R- or C-chamfered corners. The insulating plates **21** are made of ceramic with excellent thermal conductivity. The ceramic here is made of a material containing, for example, aluminum oxide, silicon nitride, or aluminum nitride as a major component.

[0034] The metal plates **22** are made of a metal having excellent thermal conductivity as a major component. Such a metal is, for example, copper, aluminum, or an alloy containing at least one of these. Plating may be applied to coat the surfaces of the metal plates **22** in order to provide improved corrosion resistance. In this case, a material used for plating is, for example, nickel, a nickel-phosphorus alloy, or a nickel-boron alloy.

[0035] The conductive circuit patterns **23** are made of a material with excellent electrical conductivity. Such a metal is, for example, copper, aluminum, or an alloy containing at least one of these. Plating may be applied to the surfaces of the conductive circuit patterns **23** to provide improved corrosion resistance. In this case, a material used for plating is, for example, nickel, a nickel-phosphorus alloy, or a nickel-boron alloy. In addition, the semiconductor chips **24** and the external connection terminals **43** to **45** are mechanically and electrically connected to the conductive circuit patterns. The conductive circuit patterns **23** may be provided as appropriate. Each has a shape to realize a desired circuit, and may also be provided in plurality.

[0036] As the insulated circuit boards **20a** to **20f** having the aforementioned components, for example, direct copper bonding (DCB) boards or active metal brazed (AMB) boards may be used. The joining members described above may also be used to join the conductive circuit patterns **23** of the insulated circuit boards **20a** to **20d** and the semiconductor chips **24** and to join the conductive circuit patterns **23** of the insulated circuit boards **20a** to **20d** and the external connection terminals **43** to **45**.

[0037] For the insulated circuit boards **20a** to **20d** on which the semiconductor chips **24** are

mounted, wiring members may mechanically and electrically connect main electrodes of the semiconductor chips **24** to each other; the main electrodes of the semiconductor chips **24** and the conductive circuit patterns **23** to each other; and the multiple conductive circuit patterns **23** to each other. The wiring members are, for example, wires and lead frames made of a material having excellent electrical conductivity. The material is, for example, gold, silver, copper, aluminum, or an alloy containing at least one of these.

[0038] The semiconductor chips **24** may be power metal-oxide-semiconductor field-effect transistors (power MOSFETs) containing silicon carbide as a major component. In the power MOSFETs, body diodes may function as free wheeling diodes (FWDs). The semiconductor chips **24** each have an input electrode (drain electrode) functioning as a main electrode on the back surface, and an output electrode (source electrode) functioning as a main electrode and control electrodes (gate electrode) on the front surface. The control electrodes may be provided at the center of one side of the front surface of the individual semiconductor chips **24**, or may be provided offset from the center along the side.

[0039] Alternatively, the semiconductor chips **24** may include a switching element containing silicon as a major component. The switching element is, for example, a reverse-conducting insulated gate bipolar transistor (RC-IGBT). The RC-IGBT is a semiconductor element having a single-chip structure with a circuit where an IGBT is connected antiparallel with a FWD. Each of the above semiconductor chips **24** has an input electrode (collector electrode) functioning as a main electrode on the back surface, and an output electrode (emitter electrode) functioning as a main electrode and control electrodes (gate electrode) on the front surface. The control electrodes may be provided at the center of one side of the front surface of each of the semiconductor chips **24**, or may be provided offset from the center along the side, as in the case of the power MOSFETs described above.

[0040] Alternatively, the semiconductor chips **24** may be, for example, semiconductor chips containing silicon as a major component and each including a switching element or a diode element. Specifically, first semiconductor chips may be switching elements while second semiconductor chips may be diode elements. The switching elements are, for example, power MOSFETs or IGBTs. Each semiconductor chip including a switching element has, for example, an input electrode (a drain electrode in the case of a power MOSFET; and a collector electrode in the case of an IGBT) functioning as a main electrode on the back surface, and a gate electrode functioning as control electrodes and an output electrode (a source electrode in the case of a power MOSFET; and an emitter electrode in the case of an IGBT) functioning as a main electrode on the front surface. As for the diode elements, for example, Schottky barrier diodes (SBD) or P-intrinsic-N (PiN) diodes are used as FWDs. Each semiconductor chip including a diode element has an output electrode (a cathode electrode) functioning as a main electrode on the back surface, and an input electrode (an anode electrode) functioning as a main electrode on the front surface.

[0041] The case **40** includes a frame portion **41**, a lid portion **42** for covering the upper part of the frame portion **41**, and the external connection terminals **43** to **45**. The outer shape of the frame portion **41** is substantially rectangular in plan view and forms a frame shape. The frame portion **41** has the housing area **41g** penetrating through the center. The frame portion **41** includes side walls **41a** to **41d** surrounding the housing area **41g** on all four sides. The side walls **41a** and **41c** correspond to the shorter sides, and the side walls **41b** and **41d** correspond to the longer sides. A bottom surface **41f** of the aforementioned frame portion **41** is joined to the outer edges of the top surface of the heat dissipation base **30** with adhesive or the like.

[0042] The external connection terminals **43** to **45** electrically connect the conductive circuit patterns **23** and an external device (not illustrated). The external connection terminals **43** to **45** are each formed of a flat plate-shaped conductive member. Part of the external connection terminals **44** and **45** is integrally held by a terminal holding portion **46**.

[0043] One end of the external connection terminal **43** is exposed on a top surface **41e** of the frame

portion **41** of the case **40**, and forms an external connection portion **43a** for connection to the external device. The external connection terminal **43** also includes horizontal portions **43b** and **43c** formed in this order toward the other end from the external connection portion **43a**. The horizontal portions **43b** and **43c** are parallel to the bottom surface of the housing area **41g**. The height of the horizontal portion **43b** is lower than that of the external connection portion **43a**, and the height of the horizontal portion **43c** is lower than that of the horizontal portion **43b**. The external connection portion **43a** and the horizontal portions **43b** and **43c** are formed, for example, by bending the flat plate-shaped external connection terminal **43**. The external connection portion **43a** and the horizontal portion **43b** as well as the horizontal portion **43b** and the horizontal portion **43c** are connected by individual coupling portions extending vertically.

[0044] The external connection portion **43a** is divided into three parts in the horizontal direction (the Y direction). A connection portion (no reference numeral given) extending downward is formed at the end of the horizontal portion **43c**. The connection portion is electrically and mechanically connected to the respective conductive circuit patterns **23** of the insulated circuit boards **20b** and **20c** via solder. The connection here may be made directly by, for example, laser or ultrasonic welding.

[0045] One end of the external connection terminal **44** is exposed at the top of the lid portion **42** of the case **40**, and forms an external connection portion **44a** for connection to the external device (not illustrated). The external connection terminal **44** includes a horizontal portion **44b** on the other end side as seen from the external connection portion **44a**. The horizontal portion **44b** is parallel to the bottom surface of the housing area **41g**, and the height of the horizontal portion **44b** is lower than that of the external connection portion **44a**. The external connection portion **44a** and the horizontal portion **44b** are formed by bending the flat plate-shaped external connection terminal **44**, and the external connection portion **44a** and the horizontal portion **44b** are connected by a coupling portion (no reference numeral given) extending vertically.

[0046] The external connection portion **44a** is divided into two parts in the horizontal direction (the Y direction). Connection portions (no reference numerals given) extending downward are formed at the end of the horizontal portion **44b**. The connection portions are electrically and mechanically connected to the respective conductive circuit patterns **23** of the insulated circuit boards **20a** and **20d** via solder. Alternatively, the connection here may be made directly by, for example, laser or ultrasonic welding.

[0047] One end of the external connection terminal **45** is exposed on the top surface **41e** of the frame portion **41** of the case **40**, and forms an external connection portion **45a** for connection to the external device. The external connection terminal **45** also includes horizontal portions **45b** and **45c** formed in this order toward the other end from the external connection portion **45a**. The horizontal portions **45b** and **45c** are parallel to the bottom surface of the housing area **41g**. The height of the horizontal portion **45b** is lower than that of the external connection portion **45a**, and the height of the horizontal portion **45c** is lower than that of the horizontal portion **45b**. The external connection portion **45a** and the horizontal portions **45b** and **45c** are formed, for example, by bending the flat plate-shaped external connection terminal **45**. The external connection portion **45a** and the horizontal portion **45b** as well as the horizontal portion **45b** and the horizontal portion **45c** are connected by individual coupling portions (no reference numerals given) extending vertically.

[0048] The external connection portion **45a** is divided into two parts in the horizontal direction (the Y direction). Connection portions (no reference numerals given) extending downward are formed at the end of the horizontal portion **45c**. The connection portions are electrically and mechanically connected to the respective conductive circuit patterns **23** of the insulated circuit boards **20b** and **20c** via solder. Alternatively, the connection here may be made directly by, for example, laser or ultrasonic welding.

[0049] The terminal holding portion **46** seals the following portions together: the coupling portion connecting the external connection portion **44a** and the horizontal portion **44b** of the external

connection terminal **44**; the coupling portions of the external connection terminal **45**, respectively connecting the external connection portion **45a** and the horizontal portion **45b** and connecting the horizontal portions **45b** and **45c**; and the horizontal portion **45b**. This maintains insulation even when the external connection terminals **44** and **45** are disposed close to each other. The terminal holding portion **46** may be made of the same material as the case **40**.

[0050] Although not illustrated in the drawings, control terminals may be provided. One ends of the control terminals are also exposed from the case **40** or the lid portion **42** of the case **40**, and the other ends are disposed in the housing area **41g**. The other ends of the control terminals are electrically connected to the control electrodes of the semiconductor chips **24** via the insulated circuit boards **20e** and **20f**. More specifically, the other ends of the control terminals are electrically and mechanically connected to conductive circuit patterns (not illustrated) of the insulated circuit boards **20e** and **20f** via solder. Alternatively, the control terminals may be directly connected by, for example, laser or ultrasonic welding. Note that the insulated circuit boards **20e** and **20f** may be mechanically and electrically connected to the control electrodes of the semiconductor chips **24** with control wires.

[0051] The horizontal portion **43c** of the external connection terminal **43**, the horizontal portion **44b** of the external connection terminal **44**, and the horizontal portion **45c** of the external connection terminal **45** are disposed at positions lower than a top surface **47a** of the sealing member **47** and are sealed with the sealing member **47**. Note that the horizontal portion **44b** of the external connection terminal **44** and the horizontal portion **45c** of the external connection terminal **45** may be covered with a resin made of a different material from that of the sealing member **47**. In this case, the top surface **47a** of the sealing member **47** may be located lower than the horizontal portion **45c** of the external connection terminal **45**. In addition, the top surface **47a** of the sealing member **47** may be located lower than the horizontal portion **43c** of the external connection terminal **43**.

[0052] The case **40** may be integrally formed by insert molding using a thermoplastic resin such that the frame portion **41** includes the external connection terminals **43** to **45**. The lid portion **42** is also molded in a similar fashion. As such a thermoplastic resin, any of the following may be used: a polyphenylene sulfide resin; a polybutylene terephthalate resin; a polybutylene succinate resin; a polyamide resin; and an acrylonitrile butadiene styrene resin.

[0053] The housing area **41g** surrounded by the heat dissipation base **30** and the frame portion **41** of the case **40** is filled with the sealing member **47**. The insulated circuit boards **20a** to **20f**, the semiconductor chips **24**, wiring members, and other components disposed in the housing area **41g** are sealed with the sealing member **47**. The sealing member **47** is an insulating polymer gel. The major component is preferably silicone gel.

[0054] Next, the heat dissipation base **30** where the insulated circuit boards **20a** to **20f** are disposed on a top surface **31e** is described with reference to FIG. **4**. FIG. **4** is a plan view of the heat dissipation base with the insulated circuit boards disposed thereon, which is included in the semiconductor module of the first embodiment. Note that the plan view of FIG. **4** depicts the heat dissipation base **30** with the insulated circuit boards **20a** to **20e** disposed thereon, excluding the case **40**, the sealing member **47**, the wires and the like from FIG. **1**.

[0055] Also in FIG. **4**, the insulated circuit boards **20a** to **20f** are represented in a simplified form. The insulated circuit boards **20a** to **20d** are rectangular in plan view, as described above. The insulated circuit boards **20a** to **20d** include lateral surfaces **20a1** to **20a4**, **20b1** to **20b4**, **20c1** to **20c4**, and **20d1** to **20d4**, respectively, that surround the four sides. The lateral surfaces **20a1**, **20a3**, **20b1**, **20b3**, **20c1**, **20c3**, **20d1**, and **20d3** correspond to the shorter sides. On the other hand, the lateral surfaces **20a2**, **20a4**, **20b2**, **20b4**, **20c2**, **20c4**, **20d2**, and **20d4** correspond to the longer sides.

[0056] The heat dissipation base **30** includes a heat dissipation plate **31** and the fixing portions **32a** and **32b**. Note that the heat dissipation plate **31** and the fixing portions **32a** and **32b** may be integrally connected, and may be made of the aforementioned metal as a major component. In

addition, the heat dissipation base **30** may have a uniform thickness throughout, including the heat dissipation plate **31** and the fixing portions **32a** and **32b**. Alternatively, the thickness of the fixing portions **32a** and **32b** may be thinner than the thickness of the heat dissipation plate **31**.

[0057] The heat dissipation plate **31** has a rectangular shape in plan view. The heat dissipation plate **31** includes the rectangular top surface **31e** and a bottom surface **31f** opposite the top surface **31e** (see FIGS. 2 and 3). The heat dissipation plate **31** further includes lateral surfaces **31a** to **31d** surrounding the top surface **31e** and the bottom surface **31f** in order on all four sides. The lateral surfaces **31a** and **31c** correspond to second edges, and the lateral surfaces **31b** and **31d** correspond to first edges. The length of the lateral surfaces **31a** and **31c** (second edges) of the heat dissipation plate **31** is denoted by length L_y , and the length of the lateral surfaces **31b** and **31d** (first edges) is denoted by length L_x . The length L_y may be about 1.0 cm, and the length L_x may be about 1.4 cm.

[0058] The four corners of the heat dissipation plate **31** are individually formed where the lateral surfaces **31a** and **31b** meet, the lateral surfaces **31b** and **31c** meet, the lateral surfaces **31c** and **31d** meet, and the lateral surfaces **31d** and **31a** meet. These corners of the heat dissipation plate **31** may be R- or C-chamfered. The heat dissipation plate **31** has fastener holes **31g** each provided at the four corners, penetrating the top surface **31e** and the bottom surface **31f** in the thickness direction (the $\pm Z$ direction). The four fastener holes **31g** are individually located, on the top surface **31e** of the heat dissipation plate **31**, within a range of distance G_x (to be described later) from the lateral surfaces **31a** and **31c** and overlap a range of distance G_y (to be described later) from the lateral surfaces **31b** and **31d**.

[0059] Center lines C_x and C_y are set on the top surface **31e**. The center line C_x is parallel to the lateral surfaces **31a** and **31c** (the paired second edges) and passes through the centers of the lateral surfaces **31b** and **31d** (the paired first edges). The center line C_y is parallel to the lateral surfaces **31b** and **31d** (the paired first edges) and passes through the centers of the lateral surfaces **31a** and **31c** (the paired second edges).

[0060] The insulated circuit boards **20a** to **20f** are disposed on the top surface **31e** of the heat dissipation plate **31**, as described above. The insulated circuit boards **20a** to **20d** are arranged in two rows and two columns on the top surface **31e** of the heat dissipation plate **31**, each separated by the center lines C_x and C_y .

[0061] The insulated circuit boards **20a** and **20b** are disposed such that the lateral surfaces **20a3** and **20b1** oppose each other at an equal distance across the center line C_x . The lateral surfaces **20a2** and **20b2** of the insulated circuit boards **20a** and **20b** are located outside (in the $+Y$ direction) of a line connecting the centers of a pair of fastener holes **31g** provided close to the lateral surface **31b** of the heat dissipation plate **31**. At this time, the shortest distance from the lateral surfaces **20a2** and **20b2** (first side) of the insulated circuit boards **20a** and **20b** to the lateral surface **31b** of the heat dissipation plate **31** is the distance G_y . Note however that the paired fastener holes **31g** formed closer to the lateral surface **31b** of the heat dissipation plate **31** are located closer to the lateral surface **31b** of the heat dissipation plate **31** than the lateral surfaces **20a2** and **20b2** of the insulated circuit boards **20a** and **20b**. That is, the lateral surfaces **20a2** and **20b2** of the insulated circuit boards **20a** and **20b** are located outside (in the $+Y$ direction) of the line connecting the centers of the paired fastener holes **31g** closer to the lateral surface **31b** of the heat dissipation plate **31**, but are not located outside (in the $+Y$ direction) of the paired fastener holes **31g**.

[0062] The insulated circuit boards **20d** and **20c** are disposed such that the lateral surfaces **20d3** and **20c1** oppose each other at an equal distance across the center line C_x . The lateral surfaces **20d4** and **20c4** of the insulated circuit boards **20d** and **20c** are located outside (in the $-Y$ direction) of a line connecting the centers of a pair of fastener holes **31g** provided close to the lateral surface **31d** of the heat dissipation plate **31**. At this time, the shortest distance from the lateral surfaces **20d4** and **20c4** (first side) of the insulated circuit boards **20d** and **20c** to the lateral surface **31d** of the heat dissipation plate **31** is also the distance G_y . Note however that the paired fastener holes **31g** formed closer to the lateral surface **31d** of the heat dissipation plate **31** are located closer to the lateral

surface **31d** of the heat dissipation plate **31** than the lateral surfaces **20d4** and **20c4** of the insulated circuit boards **20d** and **20c**. That is, the lateral surfaces **20d4** and **20c4** of the insulated circuit boards **20d** and **20c** are located outside (in the $-Y$ direction) of the line connecting the centers of the paired fastener holes **31g** closer to the lateral surface **31d** of the heat dissipation plate **31**, but are not located outside (in the $-Y$ direction) of the paired fastener holes **31g**. The distance G_y of the insulated circuit boards **20a** and **20b** and the insulated circuit boards **20d** and **20c** to the lateral surfaces **31b** and **31d** of the heat dissipation base **30** may be, for example, 5% or more and 15% or less of the length L_y .

[0063] The insulated circuit boards **20a** and **20d** are disposed such that the lateral surfaces **20a4** and **20d2** oppose each other at an equal distance across the center line C_y . At this time, the shortest distance from the lateral surfaces **20a1** and **20d1** (second side) of the insulated circuit boards **20a** and **20d** to the lateral surface **31a** of the heat dissipation plate **31** is the distance G_x . The insulated circuit boards **20b** and **20c** are disposed such that the lateral surfaces **20b4** and **20c2** oppose each other at an equal distance across the center line C_y . At this time, the shortest distance from the lateral surfaces **20b3** and **20c3** (second side) of the insulated circuit boards **20b** and **20c** to the lateral surface **31c** of the heat dissipation plate **31** is the distance G_x . The distance G_x of the insulated circuit boards **20a** and **20b** and the insulated circuit boards **20d** and **20c** to the lateral surfaces **31a** and **31c** of the heat dissipation base **30** may be, for example, 15% or more and 20% or less of the length L_x .

[0064] Furthermore, the distance G_y of the insulated circuit boards **20a** and **20b** and the insulated circuit boards **20d** and **20c** to the lateral surfaces **31b** and **31d** of the heat dissipation base **30** is shorter than the distance G_x of the insulated circuit boards **20a** and **20b** and the insulated circuit boards **20d** and **20c** to the lateral surfaces **31a** and **31c** of the heat dissipation base **30**. Note that the gaps between the respective insulated circuit boards **20a** to **20d** need only be large enough to maintain insulation.

[0065] The insulated circuit board **20e** is provided between the fastener holes **31g** on the lateral surface **31a** side of the top surface **31e** of the heat dissipation plate **31**. The insulated circuit board **20f** is provided between the fastener holes **31g** on the lateral surface **31c** side of the top surface **31e** of the heat dissipation plate **31**. The center line C_y passes through the centers of the insulated circuit boards **20e** and **20f**.

[0066] The fixing portions **32a** and **32b** are each provided in the central parts of the respective lateral surfaces **31b** and **31d** (the first edges) of the heat dissipation plate **31** in such a manner as to protrude outward. The first embodiment depicts the case where the fixing portions **32a** and **32b** are provided at the centers of the lateral surfaces **31b** and **31d** through which the center line C_x passes. The central parts of the lateral surfaces **31b** and **31d** need not necessarily be the centers of the lateral surfaces **31b** and **31d** through which the center line C_x passes, and may include a certain width. For example, the central parts may have a width of 30% or more and 70% or less of the length L_x .

[0067] Fixing holes **32a1** and **32b1** penetrate the fixing portions **32a** and **32b** in the thickness direction of the heat dissipation plate **31**. Here, the fixing holes **32a1** and **32b1** each have a concave shape recessed toward the paired lateral surfaces **31b** and **31d** (the first edges). The shapes of the concave fixing holes **32a1** and **32b1** may be line-symmetric about the center line C_x . The shapes of the fixing holes **32a1** and **32b1** are not limited to being concave, and may be circular opening holes in plan view. In this case, the fixing holes **32a1** and **32b1** may also be line-symmetric about the center line C_x . In addition, the outer corners of the fixing portions **32a** and **32b** may be tapered.

[0068] That is, the heat dissipation base **30** integrally including the heat dissipation plate **31** and the fixing portions **32a** and **32b** includes the paired opposing lateral surfaces **31b** and **31d** (the first edges), which extend along the longer-side direction of the top surface **31e** and the bottom surface **31f**, and the paired opposing lateral surfaces **31a** and **31c** (the second edges), which extend along the shorter-side direction of the top surface **31e** and the bottom surface **31f**, and further includes the

fixing portions **32a** and **32b** provided on the paired lateral surfaces **31b** and **31d** in such a manner as to protrude outward.

[0069] Next, attachment of the semiconductor module **10** including the above-described heat dissipation base **30** to a cooling module is described with reference to FIG. 5. FIG. 5 is a side view of the semiconductor module attached to a cooling module according to the first embodiment. Note that FIG. 5 omits the illustration of the case **40** of the semiconductor module **10** in FIG. 1 attached to a cooling module **50**. Thus, FIG. 5 depicts a case where the heat dissipation base **30** with the insulated circuit boards **20a** to **20d** provided thereon is attached to the cooling module **50**. The side view of FIG. 5 illustrates the semiconductor module **10** of FIG. 1 attached to the cooling module **50** as viewed in the +Y direction.

[0070] The cooling module **50** includes a top surface **51** on which the bottom surface **31f** of the heat dissipation base **30** of the semiconductor module **10** is disposed. The top surface **51** is larger in area than the bottom surface **31f** of the heat dissipation base **30** (including the fixing portions **32a** and **32b**), and is substantially flat. The cooling module **50** may be, for example, a heat radiator having heat dissipation fins or a cooling device where a refrigerant circulates the inside.

[0071] The heat dissipation base **30** of the semiconductor module **10** is disposed on the top surface **51** of the cooling module **50**. Warping occurs in the heat dissipation base **30** due to thermal changes during the manufacturing process of the semiconductor module **10**. Screws **33** (fixing members) inserted through the four fastener holes **31g** and the fixing holes **32a1** and **32b1** (FIG. 5 depicts two fastener holes **31g** and the fixing hole **32b1**) are fastened (fixed) to the top surface **51** of the cooling module **50**, and in this manner, the heat dissipation base **30** is attached to the top surface **51** of the cooling module **50**.

[0072] The bottom surface **31f** of the heat dissipation base **30** is disposed on the top surface **51** of the cooling module **50** via a joining member (not illustrated). The joining member may be, for example, a thermal interface material (hereinafter, TIM). Examples of the TIM include various materials, such as thermal conductive grease, an elastomer sheet, room temperature vulcanization (RTV) rubber, gel, and a phase change material.

[0073] Here, a reference example is described in relation to the semiconductor module **10** (the heat dissipation base **30**) of the first embodiment. A heat dissipation base **300** included in a semiconductor module of the reference example (not illustrated) is obtained by excluding the fixing portions **32a** and **32b** from the heat dissipation base **30** of the first embodiment. The semiconductor module **10** including the aforementioned heat dissipation base **300** and attached to the cooling module **50** is described with reference to FIGS. 6 and 7. FIG. 6 is a side view of the semiconductor module attached to the cooling module (at room temperature) according to the reference example. FIG. 7 is a side view of the semiconductor module attached to the cooling module (at high temperature) according to the reference example.

[0074] Note that FIGS. 6 and 7 correspond to FIG. 5, and illustrate the semiconductor module of the reference example as viewed in the +Y direction. FIGS. 6 and 7 also omit the illustration of the case **40**. Therefore, FIGS. 6 and 7 represent the case where the heat dissipation base **300** with the insulated circuit boards **20a** to **20f** provided thereon is attached to the cooling module **50**.

[0075] The heat dissipation base **300** of the semiconductor module of the reference example is also disposed on the top surface **51** of the cooling module **50** via a joining member (not illustrated). The heat dissipation base **300** is warped, as described above. The screws **33** inserted through the four fastener holes **31g** (two fastener holes **31g** are illustrated in FIGS. 6 and 7) are fastened to the top surface **51** of the cooling module **50**, and the aforementioned heat dissipation base **300** is attached to the top surface **51** of the cooling module **50**. In this manner, at room temperature when the semiconductor module of reference example is not in operation, the heat dissipation base **300** maintains the warpage immediately after the manufacture of the semiconductor module, as depicted in FIG. 6.

[0076] Next, the semiconductor module attached to the cooling module **50** represented in FIG. 6 is

operated. The semiconductor module of the reference example generates heat in response to operation, and is then cooled by the cooling module **50**. When the semiconductor module repeats such thermal changes, warping occurs in the heat dissipation base **300** (the heat dissipation plate **31**).

[0077] In the heat dissipation base **300**, the fastener holes **31g** at the four corners are fixed to the top surface **51** of the cooling module **50**. The heat dissipation base **300** warps upwardly, as illustrated in FIG. 7, also due to the difference in thermal expansion coefficient between the heat dissipation base **300** and the insulated circuit boards **20a** to **20f**. When the heat dissipation plate **31** warps upwardly, a gap is created between the bottom surface **31f** of the heat dissipation plate **31** and the top surface **51** of the cooling module **50**. As a result, the joining material flows out of the heat dissipation base **300** through the gap. This reduces the thermal conductivity of the heat dissipation base **300** to the cooling module **50**.

[0078] Note that the insulated circuit boards **20a** to **20d** (FIG. 7 depicts only the insulated circuit boards **20d** and **20c**) are joined to the top surface **31e** of the heat dissipation base **300**. When the heat dissipation base **300** is warped upward, parts of the individual insulated circuit boards **20a** to **20d**, close to the central portion of the heat dissipation base **300** in plan view, are separated from the top surface **31e** of the heat dissipation base **300**. This reduces the thermal conductivity of the insulated circuit boards **20a** to **20d** to the heat dissipation base **300**.

[0079] On the other hand, expansion of the heat dissipation base **300** may be considered to improve the heat dissipation performance. However, this hinders miniaturization of the semiconductor module. In particular, in the semiconductor module of the reference example, the insulated circuit boards **20a** to **20f** are arranged in a limited range on the top surface **31e** of the heat dissipation base **300** without expanding the top surface **31e**. The warping causes the semiconductor module of the reference example to lose heat dissipation performance, which makes the semiconductor module of the reference example more susceptible to failure and in turn reduces reliability.

[0080] On the other hand, the semiconductor module **10** of the first embodiment described above includes the multiple insulated circuit boards **20a** to **20f** and the heat dissipation base **30** including the top surface **31e** on which the multiple insulated circuit boards **20a** to **20f** are disposed and having the fastener holes **31g** each located at the corners of the top surface **31e** and penetrating the heat dissipation base **30** in the thickness direction. The heat dissipation base **30** further includes the paired fixing portions **32a** and **32b** through which the fixing holes **32a1** and **32b1** penetrate in the thickness direction. The fixing portions **32a** and **32b** are provided in such a manner as to protrude, in plan view, outward relative to the paired opposing lateral surfaces **31b** and **31d** (the first edges) running along the longer-side direction of the top surface **31e** from the central parts of the paired lateral surfaces **31b** and **31d** (the first edges). In the aforementioned semiconductor module **10**, the fixing portions **32a** and **32b** of the paired lateral surfaces **31b** and **31d** (the first edges) are fixed to the top surface **51** of the cooling module **50**, together with the fastener holes **31g** at the four corners of the heat dissipation base **30**. This suppresses the occurrence of upward convex warpage in the heat dissipation base **30**. This also suppresses the outflow of the joining material from between the heat dissipation base **30** and the top surface **51** of the cooling module **50**, which in turn suppresses detachment of the insulated circuit boards **20a** to **20f** from the heat dissipation base **30**. As a result, a decrease in the heat dissipation of the semiconductor module **10** is suppressed, which thus suppresses degradation of the reliability of the semiconductor module **10**.

[0081] The above-described semiconductor module **10** may be provided in plurality and arranged in parallel to constitute a power electronics device. The power electronics device in this case is described with reference to FIG. 8. FIG. 8 is a plan view of the power electronics device including the semiconductor modules of the first embodiment. Note that FIG. 8 omits the illustration of the cases **40** of the semiconductor modules **10**.

[0082] A power electronics device **1** includes two semiconductor modules **10** (heat dissipation bases **30**) and the cooling module **50** on which the two semiconductor modules **10** are disposed.

The two semiconductor modules **10** are adjacent to each other on the top surface **51** of the cooling module **50**, and the respective lateral surfaces (the first edges) of the adjacent semiconductor modules **10** are arranged to oppose each other. In addition, the two semiconductor modules **10** are fastened to the top surface **51** of the cooling module **50** with the screws **33** inserted through the fastener holes **31g**.

[0083] Furthermore, the two adjacent semiconductor modules **10** are fixed to the top surface **51** of the cooling module **50** with a common screw **33** inserted into an opening formed by combining the concave fixing hole **32a1** and the opposing concave fixing hole **32b1** provided on the respective first edges. The diameter of the opening corresponds to the diameter of the screw **33** inserted thereto.

[0084] Therefore, the inclusion of the fixing portions **32a** and **32b** in each semiconductor module **10** facilitates the connection of multiple semiconductor modules **10**. When connecting the semiconductor modules **10**, the opposing fixing portions **32a** and **32b** are aligned together, which allows the interval between the adjacent semiconductor modules **10** to be as narrow as possible.

[0085] Note that the shape of the fixing portions **32a** and **32b** is merely an example, and is not limited to the shape depicted in the first embodiment as long as the fixing holes **32a1** and **32b1** of the fixing portions **32a** and **32b** are aligned together to form an opening through which the screw **33** is inserted.

(b) Second Embodiment

[0086] Next, a second embodiment describes a case where two insulated circuit boards are provided in the semiconductor module **10** of the first embodiment, with reference to FIG. 9. FIG. 9 is a plan view of a semiconductor module according to the second embodiment. A case depicted in FIG. 9 differs from FIG. 4 of the first embodiment in that only the insulated circuit boards **20a** and **20b** are provided, and the size of the heat dissipation base **30** is adjusted to match the insulated circuit boards **20a** and **20b**. Note that FIG. 9 omits the illustration of the insulated circuit boards **20e** and **20f**. In addition, the insulated circuit boards **20a** and **20b** are illustrated in a simplified manner. The insulated circuit boards **20a** and **20b** are rectangular in plan view, as described above. [0087] A heat dissipation base **30a** of the second embodiment also includes the heat dissipation plate **31** and the fixing portions **32a** and **32b**. Note that the heat dissipation plate **31** and the fixing portions **32a** and **32b** may be integrally connected to each other, and may be made of the aforementioned metal as the major component.

[0088] The heat dissipation plate **31** has a rectangular shape in plan view. The heat dissipation plate **31** includes the rectangular top surface **31e** and the bottom surface **31f** (not illustrated) opposite the top surface **31e**. The heat dissipation plate **31** further includes the lateral surfaces **31a** to **31d** surrounding the top surface **31e** and the bottom surface **31f** in order on all four sides. The lateral surfaces **31a** and **31c** correspond to the second edges along the shorter-side direction while the lateral surfaces **31b** and **31d** correspond to the first edges along the longer-side direction. The length of the lateral surfaces **31a** and **31c** (the second edges) of the heat dissipation plate **31** is the length L_y , and the length of the lateral surfaces **31b** and **31d** (the first edges) is the length L_x . The length L_y may be about 0.5 cm, and the length L_x may be about 1.4 cm. Thus, the heat dissipation plate **31** of the second embodiment is different from the heat dissipation plate **31** of the first embodiment only in size, and has the same configuration. In addition, the center lines C_x and C_y are set in the same manner.

[0089] The fixing portions **32a** and **32b** also have the same shape and size as those in the first embodiment, have the fixing holes **32a1** and **32b1** provided therein, and are connected to the central parts of the lateral surfaces **31b** and **31d** of the heat dissipation plate **31** as in the first embodiment.

[0090] The: insulated circuit boards **20a** and **20b** are disposed such that the lateral surfaces **20a3** and **20b1** oppose each other at an equal distance across the center line C_x . The lateral surfaces **20a2** and **20b2** of the insulated circuit boards **20a** and **20b** are located outside (in the +Y direction) of a

line connecting the centers of a pair of fastener holes **31g** provided close to the lateral surface **31b** of the heat dissipation plate **31**. At this time, the shortest distance from the lateral surfaces **20a2** and **20b2** of the insulated circuit boards **20a** and **20b** to the lateral surface **31b** of the heat dissipation plate **31** is the distance Gy. The lateral surfaces **20a4** and **20b4** of the insulated circuit boards **20a** and **20b** are located outside (in the -Y direction) of a line connecting the centers of a pair of fastener holes **31g** provided close to the lateral surface **31d** of the heat dissipation plate **31**. At this time, the shortest distance from the lateral surfaces **20a4** and **20b4** of the insulated circuit boards **20a** and **20b** to the lateral surface **31d** of the heat dissipation plate **31** is the distance Gy. The distance Gy of the insulated circuit boards **20a** and **20b** to the lateral surfaces **31b** and **31d** of the heat dissipation base **30a** may be, for example, 5% or more and 15% or less of the length Ly, as in the first embodiment.

[0091] The insulated circuit boards **20a** and **20b** are arranged symmetrically with respect to the center line Cy. In this case, the shortest distance from the lateral surfaces **20a1** and **20b3** of the insulated circuit boards **20a** and **20b** to the lateral surfaces **31a** and **31c** of the heat dissipation plate **31** is the distance Gx. The distance Gx of the insulated circuit boards **20a** and **20b** to the lateral surfaces **31a** and **31c** of the heat dissipation base **30a** may be, for example, 15% or more and 20% or less of the length Lx, as in the first embodiment. The distance Gy of the insulated circuit boards **20a** and **20b** to the lateral surfaces **31b** and **31d** of the heat dissipation base **30a** is shorter than the distance Gx of the insulated circuit boards **20a** and **20b** to the lateral surfaces **31a** and **31c** of the heat dissipation base **30a**.

[0092] The heat dissipation base **30a** of the second embodiment also further includes the paired fixing portions **32a** and **32b** through which the fixing holes **32a1** and **32b1** penetrate in the thickness direction. The fixing portions **32a** and **32b** are provided in such a manner as to protrude, in plan view, outward relative to the paired opposing lateral surfaces **31b** and **31d** (the first edges) of the top surface **31e** from the central parts of the paired lateral surfaces **31b** and **31d** (the first edges). In the aforementioned semiconductor module **10** of the second embodiment, the fixing portions **32a** and **32b** of the paired lateral surfaces **31b** and **31d** (the first edges) are also fixed to the top surface **51** of the cooling module **50**, together with the fastener holes **31g** at the four corners of the heat dissipation base **30a**. This suppresses the occurrence of upward convex warpage in the heat dissipation base **30a**. This also suppresses the outflow of the joining material from between the heat dissipation base **30a** and the top surface **51** of the cooling module **50**, which in turn suppresses detachment of the insulated circuit boards **20a** and **20b** from the heat dissipation base **30a**. As a result, a decrease in the heat dissipation of the semiconductor module **10** is suppressed, which thus suppresses degradation of the reliability of the semiconductor module **10**.

[0093] In addition, the semiconductor modules **10** (the heat dissipation base **30a**) of the second embodiment may also be fixed to the top surface **51** of the cooling module **50** with a common screw **33** inserted into an opening formed by combining the fixing holes **32a1** and **32b1** of the fixing portions **32a** and **32b** opposing each other, as in FIG. 8.

(c) Third Embodiment

[0094] Next, a third embodiment describes a semiconductor module in which the fixing portions have a shape different from that of the first embodiment, with reference to FIG. 10. FIG. 10 is a plan view of the semiconductor module according to the third embodiment.

[0095] A heat dissipation base **30b** may be made of the same material as the heat dissipation base **30** of the first embodiment, and includes the heat dissipation plate **31** and the fixing portions **32a** and **32b**. The heat dissipation plate **31** may be the same as in the first embodiment. In addition, the insulated circuit boards **20a** to **20f** are provided on the top surface **31e** of the heat dissipation plate **31**, as in the first embodiment.

[0096] The fixing portions **32a** and **32b** are each provided in the central parts of the respective lateral surfaces **31b** and **31d** (the first edges) of the heat dissipation plate **31** in such a manner as to protrude outward. The fixing portions **32a** and **32b** may be trapezoidal in plan view. In this case,

one side of the trapezoid connecting the upper base and the lower base forms a right angle with the upper base and the lower base. Therefore, the fixing portions **32a** and **32b** are respectively shifted to the opposite sides to each other in a direction perpendicular to the center line Cx passing through the centers of the lateral surfaces **31b** and **31d** (the first edges). Furthermore, the fixing portions **32a** and **32b** are point-symmetric about the center point of the top surface **31e** of the heat dissipation plate **31**.

[0097] In addition, the fixing portions **32a** and **32b** have the fixing holes **32a1** and **32b1** each formed closer to the upper base, which is shorter than the lower base. Therefore, the fixing holes **32a1** and **32b1** are provided in the protruding parts of the fixing portions **32a** and **32b**, respectively, in such a manner that they are shifted in positions in opposite directions to each other about the center line Cx and are point-symmetric about the center point of the top surface **31e** of the heat dissipation plate **31**. In this way, the fixing portions **32a** and **32b** (the fixing holes **32a1** and **32b1**) are not aligned straight on the center line Cx, but are provided within the above-mentioned ranges in the central parts of the lateral surfaces **31b** and **31d**.

[0098] In the semiconductor module of the third embodiment, the fixing portions **32a** and **32b** of the paired lateral surfaces **31b** and **31d** (the first edges) are also fixed to the top surface **51** of the cooling module **50**, together with the fastener holes **31g** at the four corners of the heat dissipation base **30b**, as in the first embodiment. This suppresses the occurrence of upward convex warpage in the heat dissipation base **30b**, as in the first embodiment. This also suppresses the outflow of the joining material from between the heat dissipation base **30b** and the top surface **51** of the cooling module **50**, which in turn suppresses detachment of the insulated circuit boards **20a** to **20f** from the heat dissipation base **30b**. As a result, a decrease in the heat dissipation of the semiconductor module is suppressed, which thus suppresses degradation of the reliability of the semiconductor module.

[0099] Also in the third embodiment, the above-described semiconductor module may be provided in plurality and arranged in parallel to constitute a power electronics device. The power electronics device in this case is described with reference to FIG. 11. FIG. 11 is a plan view of the power electronics device including the semiconductor modules of the third embodiment. Note that FIG. 11 omits the illustration of the cases **40** of the semiconductor modules.

[0100] A power electronics device **1 a** includes two semiconductor modules (the heat dissipation bases **30b**) and the cooling module **50** similar to that of the first embodiment on which the two semiconductor modules (the heat dissipation bases **30b**) are disposed. The two semiconductor modules (the heat dissipation bases **30b**) are adjacent to each other on the top surface **51** of the cooling module **50**, and the respective lateral surfaces (the first edges) of the adjacent semiconductor modules (the heat dissipation bases **30b**) are arranged to oppose each other. In addition, the two semiconductor modules (the heat dissipation bases **30b**) are fastened to the top surface **51** of the cooling module **50** with the screws **33** inserted through the fastener holes **31g**.

[0101] In addition, the two adjacent semiconductor modules (the heat dissipation bases **30b**) are fixed to the top surface **51** of the cooling module **50** with the screws **33** inserted through the respective fixing holes **32a1** and **32b1** of the trapezoidal fixing portions **32a** and **32b** provided in the respective lateral surfaces (first edges) and fitting together in a staggered fashion.

[0102] Therefore, also in the case of the semiconductor module (the heat dissipation base **30b**) of the third embodiment, the inclusion of the fixing portions **32a** and **32b** facilitates the connection of multiple semiconductor modules (the heat dissipation bases **30b**). When connecting the semiconductor modules, the opposing fixing portions **32a** and **32b** are aligned each other, which allows the interval between the adjacent semiconductor modules (the heat dissipation bases **30b**) to be as narrow as possible.

[0103] Note that the shape of the fixing portions **32a** and **32b** of the third embodiment is merely an example, and is not limited to the shape described in the third embodiment as long as the fixing holes **32a1** and **32b1** of the fixing portions **32a** and **32b** are aligned with each other when the

semiconductor modules (the heat dissipation bases **30b**) are disposed adjacent to each other.

[0104] According to an aspect, it is possible to reduce deformation of the heat dissipation bases due to thermal changes.

[0105] All examples and conditional language provided

[0106] herein are intended for the pedagogical purposes of aiding the reader in understanding the invention and the concepts contributed by the inventor to further the art, and are not to be construed as limitations to such specifically recited examples and conditions, nor does the organization of such examples in the specification relate to as of the superiority and inferiority of the invention. Although one or more embodiments of the present invention have been described in detail, it should be understood that various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

Claims

1. A semiconductor module, comprising: a plurality of insulated circuit boards; and a heat dissipation base having a rectangular top surface on which the plurality of insulated circuit boards is disposed, and having four fastener holes that penetrate through the heat dissipation base in a thickness direction at respective ones of corners of the top surface, wherein: the heat dissipation base further includes a pair of fixing portions each having a fixing hole penetrating therethrough in the thickness direction, the pair of fixing portions protruding, in a plan view of the semiconductor module, outwardly respectively from central parts of a pair of first edges of the top surface, which oppose each other and extend in a longer-side direction of the top surface.
2. The semiconductor module according to claim 1, wherein: a shortest distance from a closest insulated circuit board among the plurality of insulated circuit boards, that is closest to one first edge of the top surface to the one first edge is shorter than a shortest distance from a closest insulated circuit board among the plurality of insulated circuit boards, that is closest to one second edge of a pair of second edges of the top surface to the one second edge, the pair of second edges opposing each other and extending in a shorter-side direction of the top surface orthogonal to the longer-side direction.
3. The semiconductor module according to claim 2, further comprising an other insulated circuit board smaller than each of the plurality of insulated circuit boards, on the top surface of the heat dissipation base at an area between the closest insulated circuit board and the one second edge that has the shortest distance.
4. The semiconductor module according to claim 2, wherein: each of the plurality of insulated circuit boards has a pair of longer-sides extending in the longer-side direction and a pair of shorter-sides extending in the shorter-side direction, the shortest distance between the closest insulated circuit board and the one first edge of the top surface is a distance between one longer-side of the closest insulated circuit board and the one first edge, and the shortest distance between the closest insulated circuit board and the one second edge of the top surface is a distance between one shorter-side of the closest insulated circuit board and the one second edge.
5. The semiconductor module according to claim 4, wherein: the plurality of insulated circuit boards is four insulated circuit boards, and are disposed on the top surface of the heat dissipation base in two rows and two columns across a center line passing through centers of the pair of first edges of the top surface and a center line passing through centers of the pair of second edges of the top surface.
6. The semiconductor module according to claim 4, wherein: the plurality of insulated circuit boards is two insulated circuit boards, and are disposed on the top surface of the heat dissipation base across a center line passing through centers of the pair of first edges of the top surface.
7. The semiconductor module according to claim 1, wherein: the fixing holes of the pair of fixing portions each have a concave shape recessed toward the pair of first edges in the plan view,

respectively.

8. The semiconductor module according to claim 1, wherein: the pair of fixing portions each protrudes in a direction away from the respective first edges and are offset from each other in opposite directions that are directions respectively away from a center line passing through centers of the pair of first edges in the plan view, and is point-symmetric about a center of the top surface of the heat dissipation base.

9. The semiconductor module according to claim 1, wherein: the pair of fixing portions is integrally formed on the pair of first edges of the heat dissipation base.

10. The semiconductor module according to claim 9, wherein: a thickness of the pair of fixing portions in the thickness direction is equal to or thinner than a thickness of the heat dissipation base.

11. The semiconductor module according to claim 1, wherein: each of the fastener holes is provided closer to a corresponding edge of the heat dissipation base than is a corresponding one of the pair of first edges of the top surface of the heat dissipation base.

12. A power electronics device, comprising: the semiconductor module of claim 7 provided in plurality; and a cooling module including a top surface on which the plurality of semiconductor modules is disposed, each of the plurality of semiconductor modules being fixed to the top surface of the cooling module with a fixing member inserted through each of the fastener holes of the heat dissipation base thereof, wherein: the plurality of semiconductor modules are disposed adjacent to each other such that one of the pair of first edges of the top surface of the heat dissipation base of one semiconductor module of an adjacent two semiconductor modules faces one of the pair of first edges of the top surface of the heat dissipation base of the other semiconductor module of the adjacent two semiconductor modules, so that the fixing hole of one of the pair of fixing portions of the one semiconductor module and the fixing hole of one of the pair of fixing portions of the other semiconductor modules penetrate therethrough in the thickness direction so that the adjacent two semiconductor modules are further fixed to the top surface of the cooling module with a fixing member inserted through the one fixing hole.

13. A power electronics device, comprising: the semiconductor module of claim 8 provided in plurality; and a cooling module including a top surface on which the plurality of semiconductor modules is disposed, each semiconductor module being fixed to the top surface of the cooling module with a fixing member inserted through each of the fastener holes of the heat dissipation base thereof, wherein: the plurality of semiconductor modules are disposed adjacent to each other such that one of the pair of first edges of the top surface of the heat dissipation base of one semiconductor module of an adjacent two semiconductor modules faces one of the pair of first edges of the top surface of the heat dissipation base of the other semiconductor module of the adjacent two semiconductor modules, so that one of the pair of fixing portions of the one semiconductor module engages with one of the pair of fixing portions of the other semiconductor modules, the adjacent two semiconductor modules being further fixed to the top surface of the cooling module with a fixing member inserted through each of the fixing holes.
